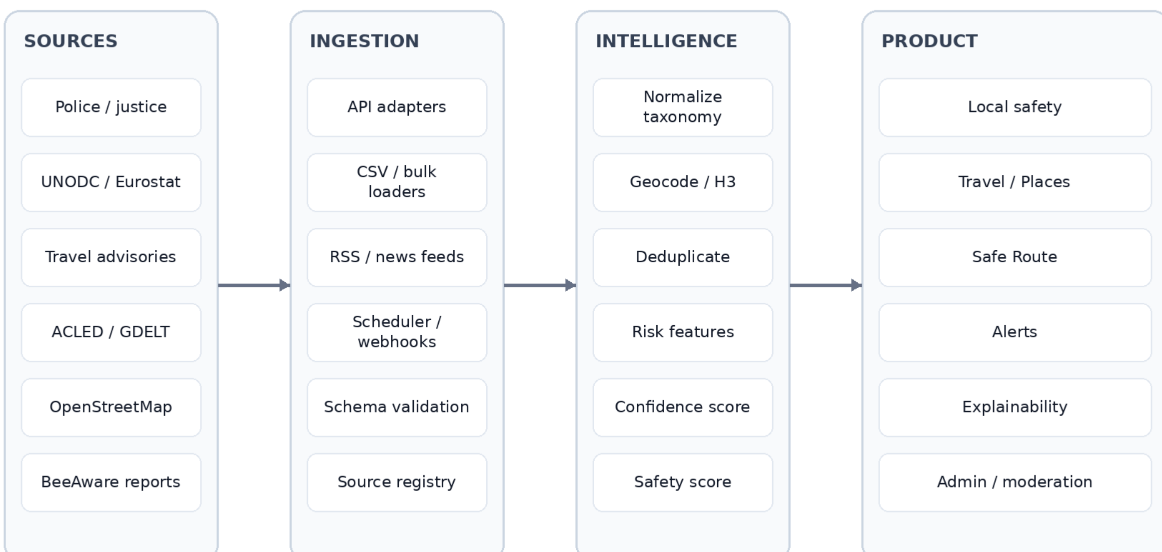
BeeAware Global

Product + Data + Engineering Blueprint

From community safety map to global location-risk intelligence

Working document for Product Owner, Engineering & Data

BeeAware Global — reference architecture



Design principle: global fallback + country/state/city adapters + explicit data-confidence.

Version 0.1 · 22 August 2026

1. Executive brief

BeeAware should evolve from an incident map into a global location-safety intelligence platform. The same data engine should support residents, travellers and route decisions, while making data quality and provenance explicit.

Product surface	Core user question	Primary output
BeeAware Local	What is the risk around me?	Neighbourhood safety, recent incidents, alerts
BeeAware Travel / Places	Where should I stay?	Place comparison, day/night and traveller risk
BeeAware Routes	How should I get there?	Fastest / Balanced / Safest route exposure
BeeAware Alerts	What changed?	Location-aware, time-sensitive risk notifications

Product principles

- Global coverage through fallbacks; local precision through country/state/city adapters.
- Never equate 'no data' with 'safe'. Data confidence is a first-class output.
- Separate observed facts, official advisories, community reports and modelled/inferred risk.
- Every score must be explainable: source, freshness, geography, transformation and confidence.
- Use a global canonical taxonomy (preferably aligned to UNODC ICCS) and map each local source into it.
- Do not expose source coordinates as exact incident addresses when the publisher anonymises or aggregates them.

2. Target user journeys

Persona	Trigger	Decision	BeeAware job
Resident	Moving / daily awareness	Is this area changing?	Baseline + recent incidents + alerts
Traveller	Choosing hotel / district	Where is safer for my party?	Compare places + traveller-specific risk
Commuter	Walking/transit journey	Which route reduces exposure?	Route risk by time of day
Parent	Family outing / school route	What is the lower-risk option?	Contextual route/place safety
Operations / moderator	Source/report review	Can this signal be trusted?	Provenance, dedupe, moderation, audit

3. Reference architecture

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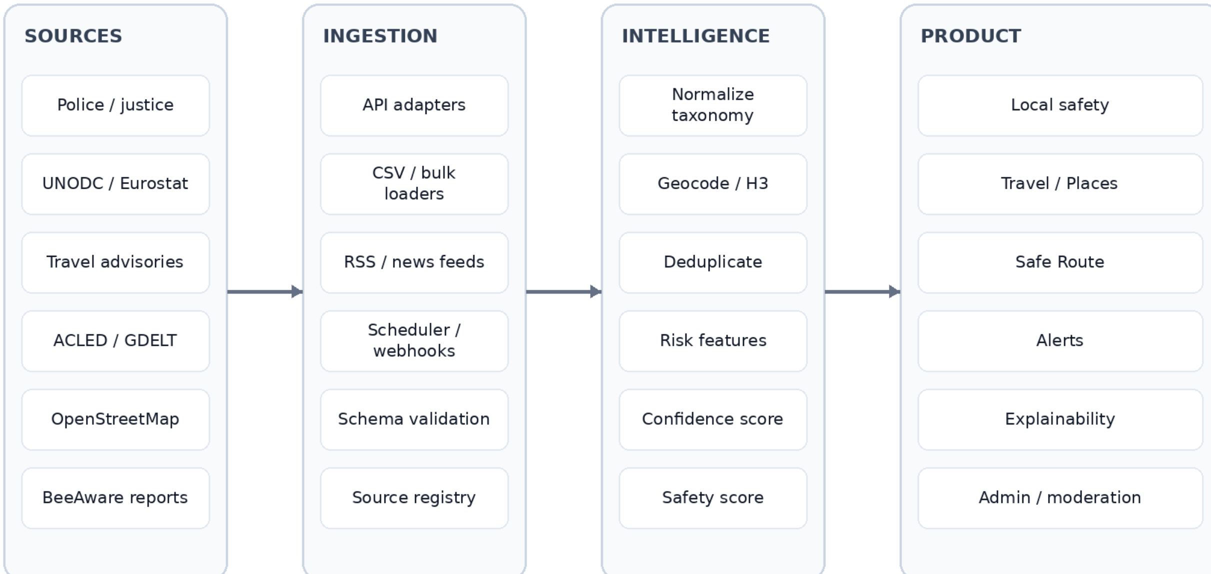
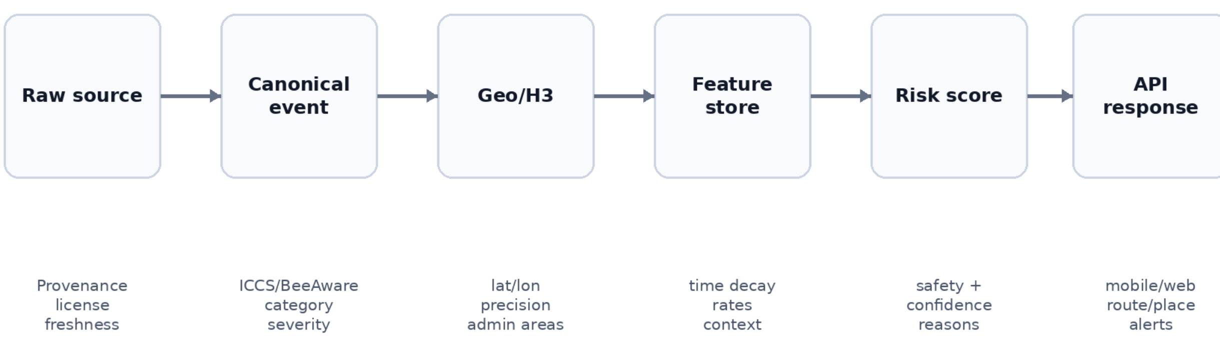


Design principle: global fallback + country/state/city adapters + explicit data-confidence.

The architecture deliberately decouples ingestion from product surfaces. A new source should be integrated once, normalised once, then become reusable across Local, Travel, Routes and Alerts.

3.1 Data flow

Event normalization & scoring pipeline



3.2 Recommended logical services

Service	Responsibility	Suggested interface

Source Registry	Metadata, licence, freshness, geo/time resolution, health	Admin + internal API
Ingestion Workers	API pulls, bulk CSV, RSS/news, retry/idempotency	Scheduled jobs / queues
Normalizer	Map local offence/event fields to canonical schema	Async worker
Geo Service	Geocode, admin boundaries, H3 indexing, precision flags	Internal API
Risk Feature Store	Precomputed time-window and context features	Postgres/PostGIS + materialized views
Risk Engine	Safety/confidence/reasons by cell/place/route	REST/Edge function
Routing Risk Layer	Scores route segments; does not replace routing provider	Route-risk API
Moderation	Community report verification / abuse / duplicates	Admin UI
Public API/BFF	Mobile/web response shaping, auth, caching	REST/GraphQL/BFF

4. Canonical data model

Entity	Minimum fields	Notes
source_registry	source_id, authority, country, type, endpoint, licence, cadence, geo_resolution, reliability	Control plane for every feed
raw_events	source_id, source_record_id, payload, ingested_at, checksum	Immutable/raw; supports replay
normalized_events	event_id, category, subcategory, occurred_at, lat/lon, geo_precision, severity, provenance	Canonical event record
geographies	geo_id, country, admin levels, geometry, H3 indexes	Boundaries and spatial joins
risk_cells	h3, window, category features, current alerts, confidence	Precomputed serving layer
travel_advisories	issuer, country/region geometry, level, text summary, effective_at	Keep issuer-specific level
community_reports	report_id, category, location, time, verification, moderation	Never merge trust state silently
places	place_id, type, point/geometry, external ids	Hotel/POI comparison surface
route_risk	route_id, segment/cell, exposure, reasons, time_context	Ephemeral or cached

4.1 Canonical event example


```json
{
 "event_id": "bea_...",
 "source_id": "uk_police",
 "source_record_id": "...",
 "country": "GB",
 "category": "THEFT",
 "iccs_code": "...",
 "occurred_at": "2026-07-01T00:00:00Z",
 "location": {"lat": 51.3, "lon": -0.3, "precision": "ANONYMISED_POINT"},
 "severity": 2,
 "verification": "OFFICIAL",
 "freshness_days": 30,
 "provenance": {"authority": "...", "licence": "..."}
}
```

5. Global source strategy

Use a layered source hierarchy. The platform should remain functional when street-level police data does not exist, but must downgrade confidence and avoid false precision.

Layer	Examples	Typical resolution	Use
Global crime baseline	UNODC; World Bank derived indicators	Country / selected subnational	Fallback and benchmarking
Regional official	Eurostat crime datasets	Regional (e.g., NUTS)	Europe fallback
National / local police	UK Police; FBI/UCR; state/city portals; SINESP; SESNSP	Street → municipality → national	Primary historical/local risk
Conflict/civil unrest	ACLED	Geocoded event	Dynamic political/civil risk
News signal	GDELT / curated publishers	Geocoded or extracted	Near-real-time corroborating signal
Travel advisory	FCDO; US State; Smartraveller; Canada	Country / region	Traveller risk and exclusion polygons
Environment/context	OpenStreetMap / public POIs	Point/line/polygon	Transport, police, hospitals, street context
Community	BeeAware reports	User-selected location	Fresh local signal with trust state

5.1 Priority source/adaptor matrix

Geo	Source	Access	Resolution	BeeAware use	Priority	Official entry point
Global	UNODC Data Portal	Official datasets/downloads	Country + selected detail	Crime baseline/taxonomy	P0	https://data.unodc.org/
UK	data.police.uk	REST API	Anonymised street points	Local incidents/outcomes	P0	https://data.police.uk/docs/
Brazil	MJSP / SINESP VDE	Bulk/open datasets	National/state; some municipal historical	Baseline + state adapters	P0	https://www.gov.br/mj/pt-br/assuntos/sua-seguranca/seguranca-publica/estatistica
USA	FBI Crime Data Explorer/UCR	API/download	Agency/state/national; city portals separately	US baseline + agency trends	P1	https://cde.ucr.cjis.gov/
Canada	Statistics Canada	Tables/geospatial explorer	Multiple geographies	CSI/crime rates/homicide etc.	P1	https://www.statcan.gc.ca/en/subjects-start/crime_and_justice
Mexico	SESNSP / datos.gob.mx	CSV/open data	State/municipality	Municipal crime trends	P1	https://www.datos.gob.mx/busca/dataset?tags=incidencia-delictiva
Colombia	datos.gov.co / Policía	Socrata/open datasets	Department/municipality; dataset-dependent	Crime + traveller-victim datasets	P1	https://www.datos.gov.co/
France	SSMSI / data.gouv.fr	Bulk open data	Commune/department/region	Local crime trends	P1	https://www.data.gouv.fr/
Europe	Eurostat	API/download	Country/regional datasets	Fallback across Europe	P1	https://ec.europa.eu/eurostat/web/crime
Australia	ABS + state sources e.g. NSW BOCSAR	Bulk/open datasets	National/state/suburb varies	Crime + daily/local where available	P1	https://bocsar.nsw.gov.au/statistics-dashboards/open-datasets.html
New Zealand	NZ Police / policedata.nz	Open data/visualisation	Area-dependent	Recorded victims/offenders/activity	P1	https://www.police.govt.nz/about-us/publications-statistics/data-and-statistics
Global	ACLED	Authenticated API	Geocoded event	Conflict/protest/civil risk	P1	https://acleddata.com/acled-api-documentation
Global	GDELT	Public APIs/data	News-derived geo	Recent signal / corroboration	P2	https://www.gdeltproject.org/
Global travel	UK FCDO	Web/data feed investigation required	Country/region	Travel advice	P1	https://www.gov.uk/foreign-travel-advice
Global	US State	Web/data	Country/region	Travel advisories	P1	https://travel.state.gov/content/travel/en/traveladvis

travel	Department	feed investigation required				ories/traveladvisories.html
Global travel	Smartraveller	Web/data feed investigation required	Country/region	Travel advisories	P1	https://www.smartraveller.gov.au/destinations
Global travel	Government of Canada	Web/data feed investigation required	Country/region	Travel advisories	P1	https://travel.gc.ca/travelling/advisories
Global context	OpenStreetMap	Overpass/planet extracts/providers	Global geo	POIs/network/context	P1	https://wiki.openstreetmap.org/wiki/Overpass_API

Engineering note: 'official entry point' is not a guarantee of a stable production API. Before implementation, validate terms/licence, rate limits, bulk-download availability, authentication, update cadence and redistribution rights. Prefer bulk/open-data feeds for scheduled ingestion where available.

6. Risk model v1

Start interpretable. Avoid an opaque ML score until the product has enough observed outcomes and calibration data.

Dimension	Example features	Output
Personal safety	violent crime rate, severity, recent violent events	0–100 subscore
Property safety	theft, robbery, burglary, vehicle crime	0–100 subscore
Current risk	7/30-day events, alerts, civil unrest, verified community reports	0–100 modifier
Traveller risk	consular levels, tourist-targeted crime, tourist-zone patterns	0–100 subscore
Mobility risk	route cell exposure, time of day, recent route-adjacent events	Route exposure
Confidence	source grade, geo precision, freshness, coverage, cross-source agreement	Very low → Very high

6.1 Suggested scoring shape

SafetyScore = 100 − clamp(BaseRisk × RecencyModifier × TimeModifier × ContextModifier, 0, 100). Keep weights configurable by country and category. Store the feature contribution vector so the UI can explain why a score moved.

Confidence should be calculated separately from Safety. A high safety score with low confidence must never be displayed as equivalent to a high score backed by recent granular official data.

6.2 Source grades

Grade	Example	Product treatment
A+	Official geocoded event with strong freshness	Highest evidence
A	Official municipality/district data updated regularly	Strong evidence
B	Regional official data	Useful baseline
C	National official/global baseline	Low local precision
D	Modelled/inferred signal	Must be labelled as inferred
U	Community report	Show verification/moderation state

7. Safe Route design

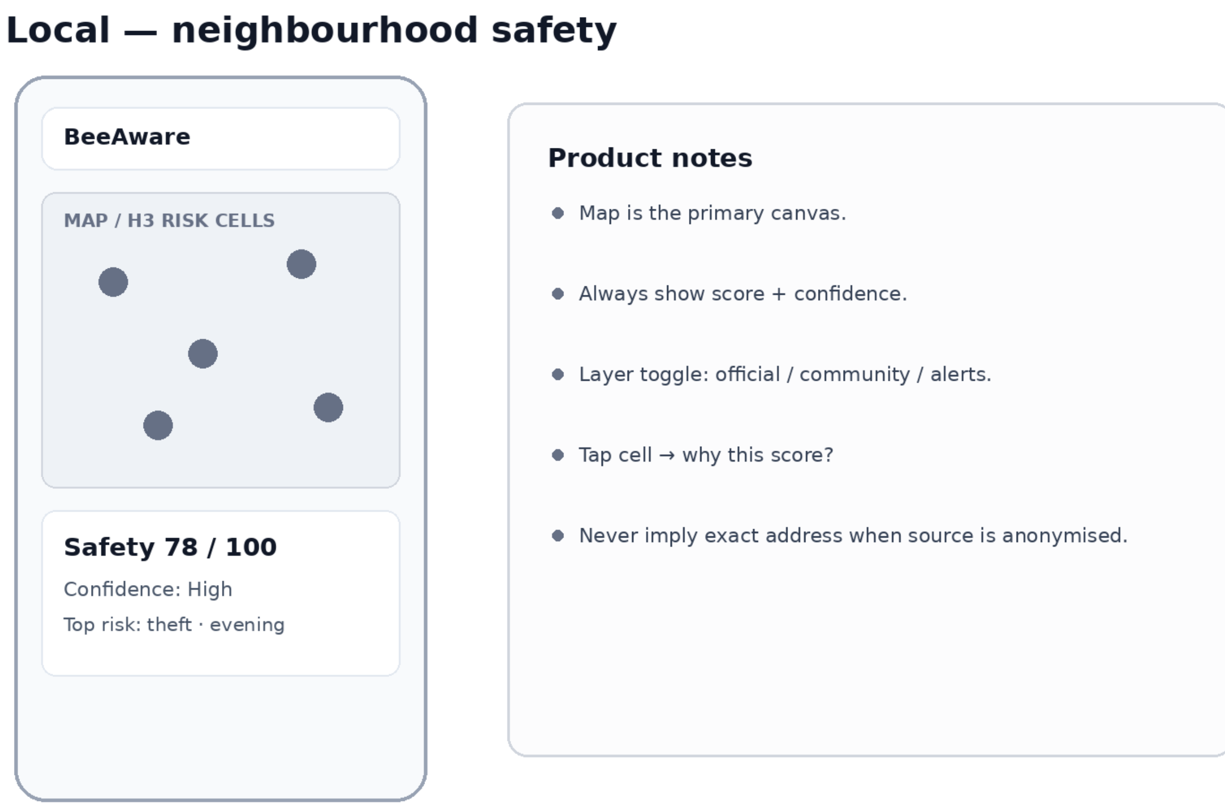
The routing provider generates candidate routes; BeeAware scores exposure. Do not build a routing engine in v1.

Step	Implementation
1. Candidate routes	Request 2–3 alternatives from routing provider
2. Segment route	Sample polyline / map to H3 cells
3. Time context	Estimate arrival time per segment and apply day/night/event modifiers
4. Risk lookup	Fetch risk features for intersected cells
5. Exposure	Σ(segment distance × segment risk × time modifier)
6. Rank	Fastest / Balanced / Safest
7. Explain	Show extra minutes, relative exposure reduction and top reasons

H3 is a strong candidate for a consistent global spatial index; validate the chosen resolution by urban density and source precision before fixing it in the schema.

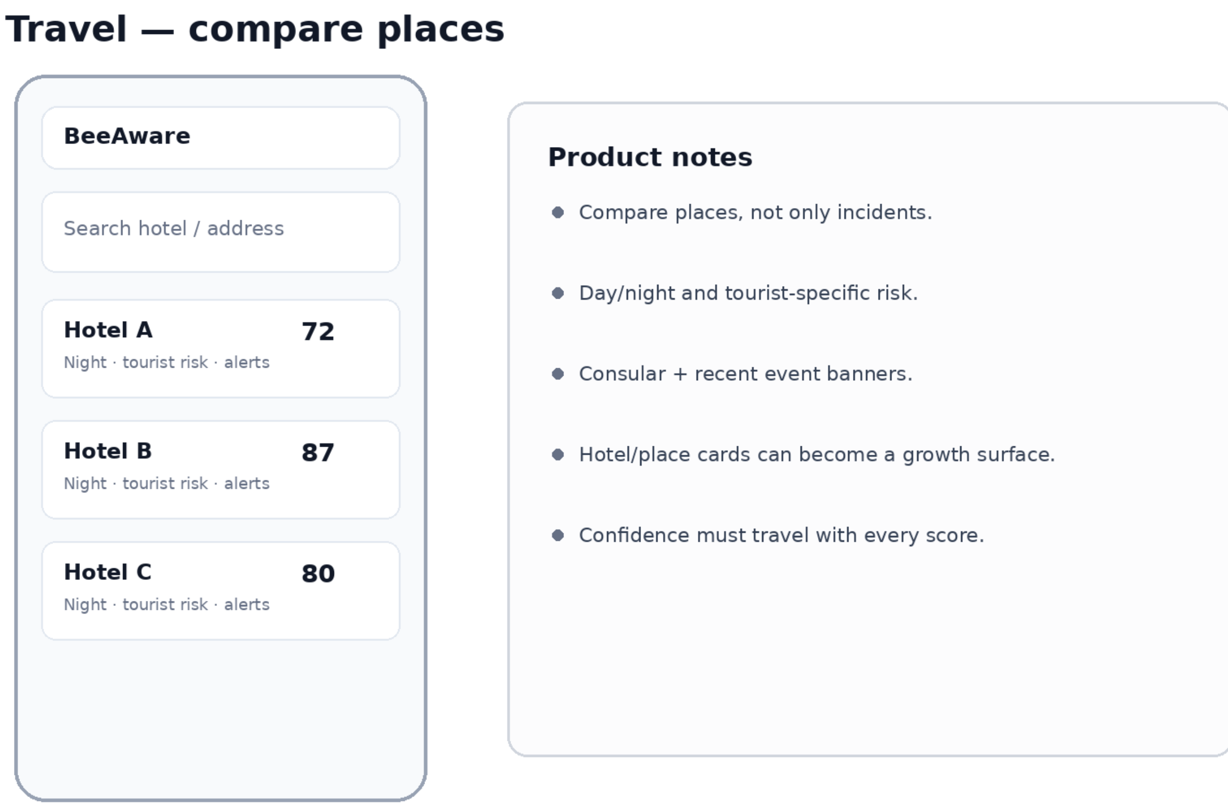
8. Wireframes

8.1 Local — neighbourhood safety



Low-fidelity wireframe for product/engineering alignment; visual design is intentionally neutral.

8.2 Travel — compare places



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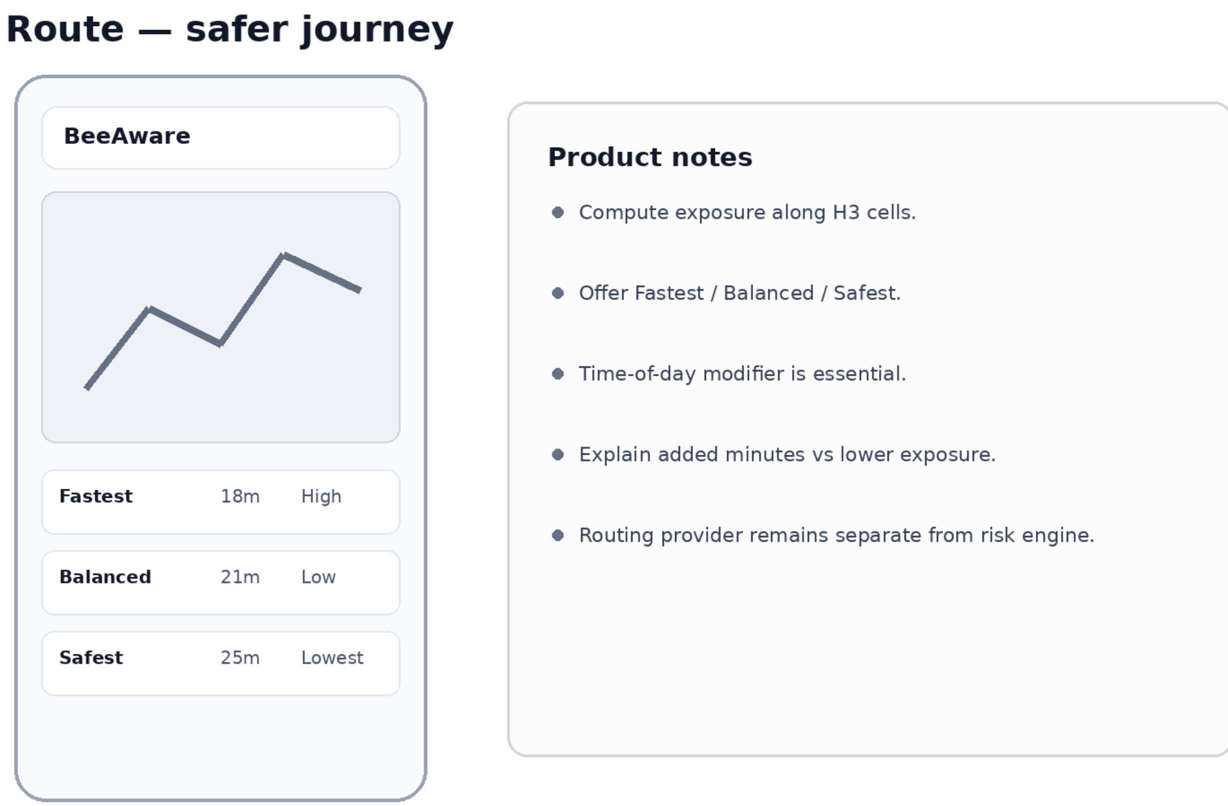
8.3 Route — safer journey



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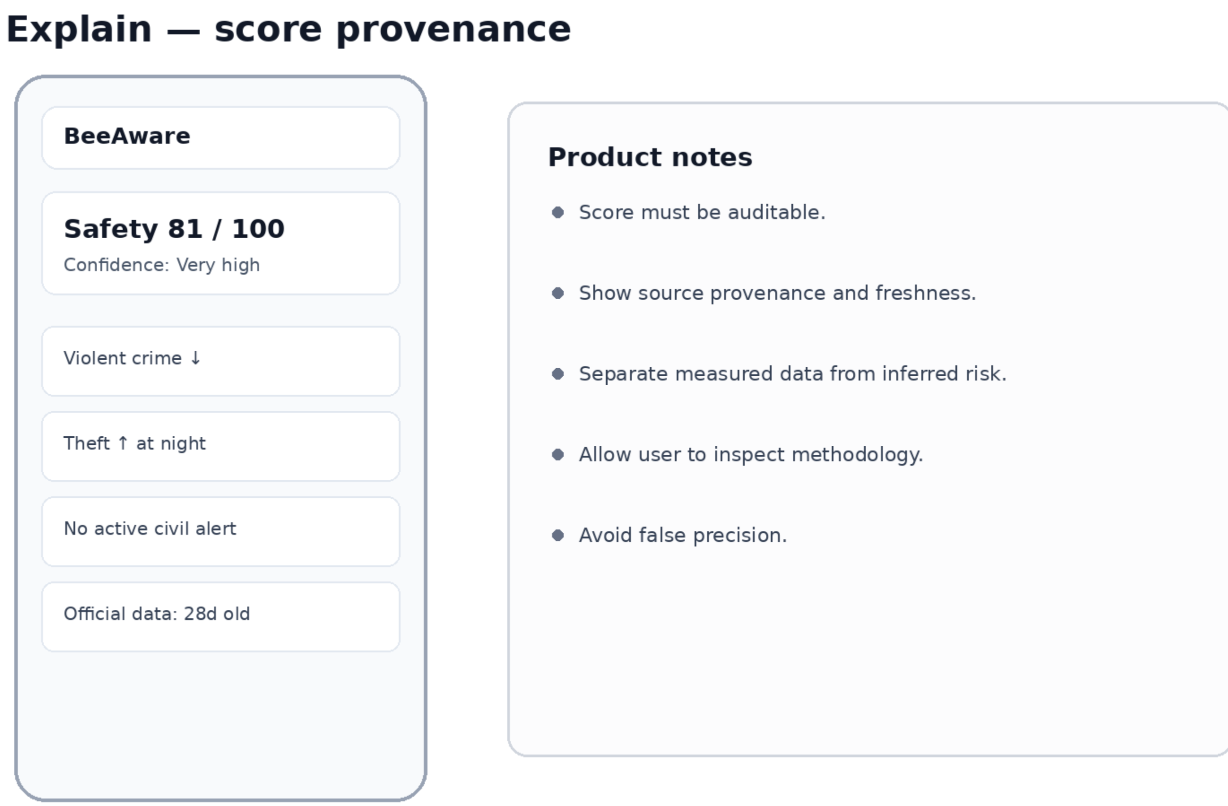
8.4 Explainability — why this score?



Low-fidelity wireframe for product/engineering alignment; visual design is intentionally neutral.

9. API contract sketch

Endpoint	Purpose	Key response
GET /v1/safety/point?lat=&lon=&at=	Safety around a point	score, confidence, reasons, coverage, source freshness
GET /v1/safety/area/{geo_id}	Area profile	subscores, trend, top categories, advisories
POST /v1/places/compare	Compare hotels/POIs/addresses	ranked places + day/night/traveller dimensions
POST /v1/routes/score	Score candidate route polylines	exposure, relative rank, risky segments, reasons
GET /v1/events?bbox=&from=&to=&category=	Map events	normalised events with provenance/precision
GET /v1/advisories?country=&bbox=	Travel/civil advisories	issuer, level, region, effective date
POST /v1/community/reports	Create report	report id + moderation/verification state
GET /v1/sources/coverage?lat=&lon=	Data transparency	available layers, grade, freshness, geo resolution

9.1 Example point response


```json
{
 "location": {"lat": 40.4168, "lon": -3.7038},
 "safety_score": 82,
 "confidence": {"level": "HIGH", "score": 0.86},
 "dimensions": {
  "personal": 85, "property": 74, "current": 90, "traveller": 81
 },
 "reasons": [
  {"factor": "PROPERTY_THEFT", "direction": "NEGATIVE", "weight": 0.22},
  {"factor": "NO_ACTIVE_CIVIL_ALERT", "direction": "POSITIVE", "weight": 0.12}
 ],
 "coverage": [
  {"layer": "OFFICIAL_CRIME", "grade": "A", "freshness_days": 31},
  {"layer": "TRAVEL_ADVISORY", "grade": "B", "freshness_days": 4}
 ]
}
```


10. Product backlog / delivery roadmap

Phase	Outcome	Engineering deliverables	PO deliverables
0 — Foundation	Global-ready data model	source_registry; raw/normalised events; ICCS mapping; provenance; confidence	taxonomy, score principles, source policy
1 — Global baseline	Useful coverage beyond current countries	UNODC + travel advisory ingestion; coverage endpoint; country fallback	global coverage UX; 'limited data' states
2 — Priority adapters	High-quality markets	UK/Brazil hardening; US/Canada/Mexico/Colombia/France/Australia/NZ adapters	market priority + acceptance criteria
3 — Travel / Places	Hotel/address comparison	places compare API; day/night; traveller dimensions	Travel journey, ranking rules, copy
4 — Dynamic risk	Recent non-police signals	ACLED; GDELT/news classifier; dedupe/corroboration	alert thresholds; false-positive policy
5 — Safe Route	Route exposure ranking	routing integration; H3 segment scoring; caching	Fastest/Balanced/Safest UX
6 — Learning loop	Calibrated risk	evaluation set, calibration, drift/coverage monitoring	metric framework and trust KPIs

10.1 Definition of Done for a new country adapter

- Source authority and licence recorded in source_registry.
- Update cadence and failure/retry behaviour documented.
- At least 95% of ingested rows pass schema validation or are quarantined with reason.

- Local offence taxonomy mapped to canonical BeeAware/ICCS categories.
- Geographic precision explicitly encoded; no fabricated coordinates.
- Freshness and coverage metrics exposed to product.
- Deduplication strategy tested on a representative sample.
- Data-quality checks cover missing periods, sudden count shifts and schema drift.
- Product fallback verified when the adapter is unavailable.
- Source attribution text approved for UI.

11. Non-functional requirements

Area	Requirement
Privacy	Do not expose PII; minimise community-report precision where necessary; support deletion/moderation workflows.
Safety/trust	No 'safe/unsafe' absolute claims; scores are comparative indicators with evidence/confidence.
Licensing	Track source licence and redistribution restrictions before public display.
Reliability	Idempotent ingestion, retries, dead-letter/quarantine, source health dashboard.
Observability	Freshness, row counts, schema failures, coverage gaps, score shifts, adapter latency.
Performance	Precompute H3 risk cells; cache point/place queries; keep route scoring bounded.
Internationalisation	Country taxonomy mappings, local language labels, timezone-aware timestamps.
Auditability	Version score formula, weights, parser and source snapshot used for each result.

12. Key product metrics

- Coverage: % of active-user locations with ≥B confidence and ≥2 independent layers.
- Freshness: median age of the best official/local source by market.
- Trust: % of score views that open 'why this score?' and report usefulness.
- Travel conversion: place comparisons → saved place / route / return visit.
- Route utility: % choosing Balanced/Safest and median added minutes accepted.
- Signal quality: precision of news/community alerts after moderation/corroboration.
- Data health: adapter success rate, schema-drift incidents and stale-source hours.

13. Decisions needed from Product + Engineering

- Choose canonical taxonomy and confirm ICCS alignment depth.
- Choose H3 resolutions and rules for sparse vs dense areas.
- Define Safety Score v1 weights and what can/cannot influence it.
- Define Confidence Score and minimum evidence required to show a numeric score.
- Select routing provider and place-search provider; keep them replaceable behind interfaces.
- Set first expansion markets after Brazil/UK based on data quality + user opportunity.
- Define advisory/news ingestion legal review and redistribution policy.
- Agree on community-report verification states and moderation SLAs.

14. Source references for implementation

UNODC Data Portal: https://data.unodc.org/

UK Police API documentation: https://data.police.uk/docs/

Brazil MJSP security statistics / SINESP: https://www.gov.br/mj/pt-br/assuntos/sua-seguranca/seguranca-publica/estatistica

FBI Crime Data Explorer: https://cde.ucr.cjis.gov/

Statistics Canada — Crime and justice: https://www.statcan.gc.ca/en/subjects-start/crime_and_justice

Mexico open data portal: https://www.datos.gob.mx/

Colombia open data portal: https://www.datos.gov.co/

France open data portal: https://www.data.gouv.fr/

Eurostat Crime: https://ec.europa.eu/eurostat/web/crime

NSW BOCSAR open datasets: https://bocsar.nsw.gov.au/statistics-dashboards/open-datasets.html

New Zealand Police data & statistics: https://www.police.govt.nz/about-us/publications-statistics/data-and-statistics

ACLED API documentation: https://acleddata.com/acled-api-documentation

GDELT Project: https://www.gdeltproject.org/

UK FCDO foreign travel advice: https://www.gov.uk/foreign-travel-advice

US State Department travel advisories: https://travel.state.gov/content/travel/en/traveladvisories/traveladvisories.html

Australia Smartraveller: https://www.smartraveller.gov.au/destinations

Canada travel advice: https://travel.gc.ca/travelling/advisories

OpenStreetMap Overpass API: https://wiki.openstreetmap.org/wiki/Overpass_API

H3 documentation: https://h3geo.org/docs/

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